Choosing an orthogonal torus element to separate the triality modules

Human Compression project

This draft supplies the triality interface in plus dimension eight. It awaits distinct review and does not classify all maximal overgroups.

Let $G = P\Omega_8^+(q)$, with inner-diagonal automorphism group J. We retain the standard automorphism structure: modulo J, the diagram group S_3 commutes with the cyclic field group. The three diagram points are the natural and two half-spin eight-dimensional projective representations. The stabilizer of one point acts by natural projective conformal maps in that representation. The Clifford and triality descriptions work in characteristic two as well; see [1, Sections 3.1 and 3.4], in particular Theorem 3.12 and the isomorphisms (3.20)–(3.21). These are borrowed structural statements.

Put $A = q^3 + 1$, $B = q + 1$, $b = q^2 - q + 1$, so $A = Bb$. Choose g of order A in the norm-one torus of $\mathbb{F}_{q^6}/\mathbb{F}_{q^3}$. On negative spaces of dimensions six and two use primitive multipliers g and $h = g^{ub}$, with u a unit modulo B. Let X be their direct sum and x its image in G. The accepted torus proof gives $X \in \Omega_8^+(q)$, $|x| = A/\gcd(2, q-1)$, and the connected adjoint centralizer

$$C_J(x) = T, \qquad |T| = AB. \tag{1}$$

In particular its order is not replaced by $\gcd(2, q-1)|x|$.

Lemma 1. *Modulo scalar signs, the natural eigenvalue exponents relative to g are $\pm 1, \pm q, \pm q^2, \pm ub$. The two half-spin lists are*

$$c_\epsilon q^j \ (0 \le j < 6), \quad d_\epsilon, -d_\epsilon, \qquad c_\epsilon = \frac{1 + q + q^2 + \epsilon ub}{2}, \quad d_\epsilon = \frac{(1 - \epsilon u)b}{2}, \quad \epsilon = \pm 1. \tag{2}$$

For even q, division by two is modulo A. For odd q the numerators are even integers, and comparison modulo scalar signs is modulo $A/2$. Changing a Spin lift changes only a scalar sign.

Proof. The Clifford construction on the exterior algebra of an isotropic four-space gives the weights $\prod z_i^{s_i/2}$, $s_i = \pm 1$, when the natural eigenvalues are z_i, z_i^{-1}. The two parities of the number of minus signs give the two half-spin modules. Indeed the ith torus factor acts with weight $z_i^{1/2}$ on exterior monomials containing the ith basis vector and with weight $z_i^{-1/2}$ on the others. Conjugation on the creation and contraction operators then has the required weights z_i, z_i^{-1}. Thus here the weights are the half-sums of $1, q, q^2, ub$ in the two parities.

Use $q^3 = -1$ and $qb = -b$ modulo A. Successively multiplying the all-positive half-sum by q gives six even-parity half-sums; the other two are $\pm(1 - u)b/2$. Starting with $1 + q + q^2 - ub$ gives the odd-parity list. For odd q the doubled identities modulo A prove the desired identities modulo $A/2$. For even q, two is invertible modulo A. □

For $q \ge 3$, the order of q modulo b is six. It divides six because $q^3 = -1$; $b > q + 1 > 2$ excludes orders one and three, and $q^2 = 1$ would force $q = 2 \pmod b$, excluding order two. Moreover

$$c_\epsilon \equiv q \pmod b. \tag{3}$$

Here $1 + q + q^2 \equiv 2q \pmod b$ and b is odd.

Proposition 2. *Suppose q is even and $q \geq 4$, or q is odd and $q \geq 7$. Choose a unit $u \pmod B$ other than 1 and -1. Every involutory automorphism of G inverting x belongs to J. There are at most AB such automorphisms. Furthermore, x belongs to no triality image of a stabilizer of a nonsingular line in a natural eight-dimensional module.*

Proof. For even q take $u = 2$. For odd $q \geq 7$, B is even and at least eight, and $\varphi(B) > 2$. The prime-power formula for Euler's function shows that $\varphi(n) = 2$ only for $n = 3, 4, 6$.

Put $M = A$ for even q and $M = A/2$ for odd q. By (3), the six exponents $c_\epsilon q^j$ have period six already modulo b; the two exponents $\pm d_\epsilon$ are divisible by b and cannot join that orbit. A similarity up to scalar signs between a half-spin and the natural spectrum must therefore match their six-element orbits. It implies $c_\epsilon = q^j \pmod M$ for some j. Reduction modulo b forces $j = 1 \pmod 6$. Since

$$2(c_\epsilon - q) = (1 + \epsilon u)b,$$

the congruence implies $u = -\epsilon \pmod B$, a contradiction. Taking inverses does not change either spectrum, so this also excludes a diagram transposition moving the natural module for an inverter.

A nontrivial involutory field projection has $q = p^f$ with even f and exponent $f/2$. If its diagram part moves the natural module, the same comparison, reduced modulo b, gives $p^{f/2}q = q^j$ after absorbing inversion into the six-element orbit. Squaring gives $q^3 = q^{2j}$, impossible for an element of order six. Ordinary natural field inverters were excluded in the acquired torus proof. An involution cannot project to a three-cycle in S_3.

The remaining diagram transposition fixes the natural module. Apply q^3-Frobenius on the six-space and q-Frobenius on the two-space. This is an involutory isometry inverting X. It is proper: its determinant in odd characteristic is $(-1)^3(-1) = 1$, and its Dickson invariant in characteristic two is $3 + 1 = 0 \pmod 2$. It therefore represents an element of J. Any other natural inverter differs from it by an element of the natural projective centralizer, the connected torus T. Thus no improper natural inverter exists. All inverters lie in J, where their pairwise quotients centralize x, proving the bound $|T|$.

Finally the natural module has no scalar-sign eigenvalue. In either half-spin module a large-orbit exponent is nonzero modulo b. A small exponent is zero modulo M only if $u = \epsilon \pmod B$, also excluded. There is no scalar-sign eigenvalue in any module. A nonsingular line fixed by a projective isometry has eigenvalue 1 or -1 on an isometric lift, since its nonzero quadratic value is preserved. Such a line therefore does not exist. □

Proposition 3. *For $q = 5$ take $u = 1$. At most $2AB$ involutory automorphisms invert x. Their fixed subgroups in G each have at most $24q^8$ elements of square one, including the identity. There are at most six triality images of nonsingular-line stabilizers containing x.*

Proof. Here $A = 126$, $B = 6$, $b = 21$, and $M = 63$. Formula (2) gives one half-spin spectrum identical modulo signs to the natural one. The other has large-orbit representative 26, of order 63, and a scalar two-space. The natural and first half-spin modules have no scalar-sign eigenvalue, so the remaining module is intrinsically distinguished. An inverter must fix its diagram point. There are at most two possible diagram projections and no field automorphism. Each such coset of J has at most $|T|$ inverters.

Use the distinguished module to represent an inverter by a conformal map a. Its six-space has irreducible eigenvalue order 63. Projective inversion has scalar 1: the degree argument forces the field exponent to be three, already giving exact inversion. On that block, the square and conformal multiplier of a are the same scalar c, as in the natural torus proof. Since its projective action is involutory, $a^2 = cI$ on the entire module.

If c is a square, the six-space contributes eigenspace dimensions three and three. The two-space adds $2, 0$, $1, 1$, or $0, 2$. Thus the full pair is $(5, 3)$, $(4, 4)$, or $(3, 5)$. The written odd-characteristic projective fixed-group bounds give $18q^8$ in the unequal cases and $24q^8$ in the

balanced case. If c is nonsquare, the two eigenspaces over the algebraic closure each have dimension four. The same bound applies, including the quadratic-field form and projective kernel.

Only the distinguished module can have a fixed nonsingular line. Its scalar two-space has minus type: the irreducible six-space is negative and the eight-space is positive. Its $q + 1 = 6$ lines are all nonsingular, and they are the only possible lines. There is at most one stabilizer for each line. This counts actual stabilizers, without replacing their multiplicity by the number of conjugacy types. □

Attribution and limits. The three representations and triality action, finite automorphism structure and Clifford construction are prior theory. The exponent calculation, primitive choice separating spectra, and resulting coset and actual-line bounds are the arguments here. The cases $q = 2, 3$ have acquired finite witnesses. The separate maximal-overgroup cover and final generation inequality remain necessary; no whole-original or new-conjecture claim is made.

References

[1] D. Barry and J.-P. Tignol, *Trialitarian triples*, arXiv:2301.10130v1, Sections 3.1–3.4. This auxiliary source does not enlarge the fixed original catalogue.

[2] D. Bubboloni, P. Spiga and Th. Weigel, *Normal* 2*-coverings of the finite simple groups and their generalizations*, arXiv:2208.08756v1, Section 8, physical pages 81–82 for the triality-related orthogonal line stabilizers.